

SAVANNA

SENTINEL



User Manual

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David van Rooijen

Dandré Nel

Daniel Cohen

Stephan Kritzinger

Michael Koch

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Roles

Savanna Sentinel has four account types, and what you can do depends on which one you have:

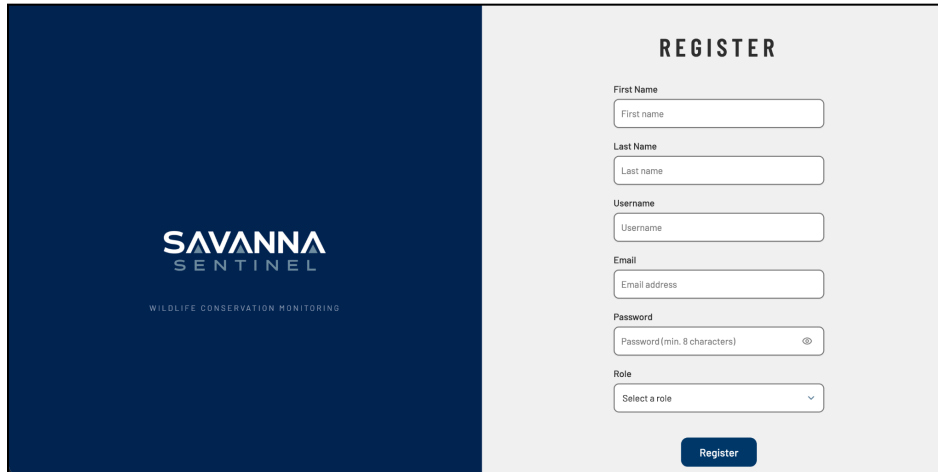
Ranger	Log sightings and incidents, request patrol routes
Analyst	Upload historical data
Community Liaison	Log sightings and incidents, gather information about events from the community.
Admin	Everything above, plus manage user accounts and view audit logs

If you are not sure which type your account is, check the top of the Profile page after logging in - it shows next to your name.

Getting Started

Creating an account

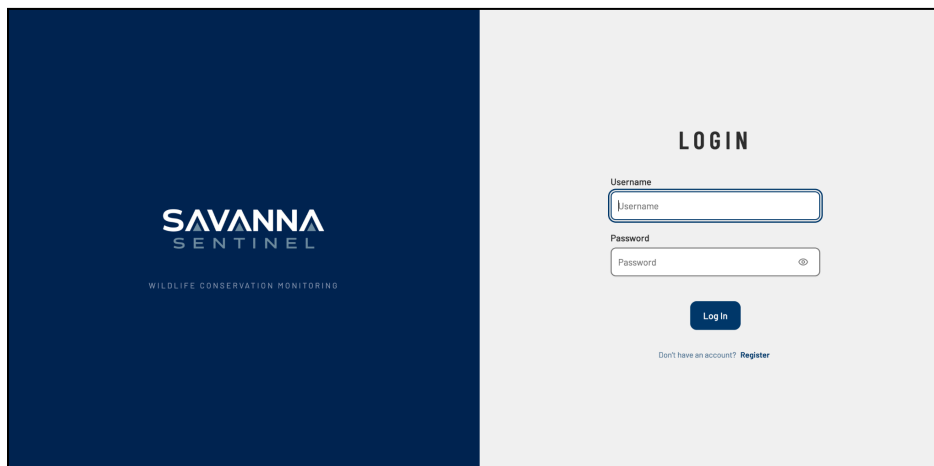
Go to the Register page and fill in your name, email, a password, and pick a role.

The screenshot shows the 'REGISTER' page of the Savanna Sentinel application. On the left is a dark blue sidebar with the 'SAVANNA SENTINEL' logo and the tagline 'WILDLIFE CONSERVATION MONITORING'. The main content area is light gray and contains a registration form. The form has fields for 'First Name', 'Last Name', 'Username', 'Email' (with 'Email address' placeholder), 'Password' (with 'Password (min. 8 characters)' placeholder and a strength indicator), and a 'Role' dropdown menu. A 'Register' button is at the bottom of the form.

After you register, your account is created but held inactive until an admin approves it.

Login

Enter your email and password on the Login page. If your account hasn't been approved yet, or if either field is wrong, you will see the same error and won't be logged in.

The screenshot shows the 'LOGIN' page of the Savanna Sentinel application. It features the same dark blue sidebar with the 'SAVANNA SENTINEL' logo and tagline. The main content area is light gray and contains a login form with fields for 'Username' and 'Password' (with a strength indicator). A 'Log In' button is positioned below the password field. At the bottom of the form, there is a link that says 'Don't have an account? Register'.

Once you are in, you will be redirected to the dashboard (otherwise tip-offs page for community members).

Your Profile

Your profile page shows your name, email, and role. From here you can:

- Update your display name
- Change your password

Profile

Profile Details

First name

Last name

Save Reset

Change Password

Current password

New password

New password must be at least 8 characters.

Confirm password

Change Password

Changing your password signs you out of any other devices you are logged into, so after changing it, you will need to log back in.

Field Reports

Rangers can log two kinds of field reports: sightings, and incidents.

Viewing your reports

The Reports page lists everything you have logged, newest first. You can filter by type, date, or location.

Field Reports

NEW REPORT **ALL REPORTS**

Filters

Type	Description	DETAILS	LOCATION	Occurred At	Submitted By	SYNC STATUS
Sighting	Lion sighted during routine observation.	Lion x5	-24.0895, 31.2001	Aug 13, 2026, 12:52 AM	ranger4	Synced
Incident	Unauthorized vehicle tracks were found inside t...	Medium	-24.2857, 31.1283	Jun 10, 2026, 12:09 AM	ranger10	Synced
Incident	Unauthorized vehicle tracks were found inside t...	Medium	-24.1801, 31.1492	Jan 6, 2026, 4:28 AM	ranger7	Synced
Incident	An animal carcass was discovered showing sign...	High	-24.2493, 31.0691	Feb 23, 2026, 10:28 AM	ranger6	Synced
Incident	An animal carcass was discovered showing sign...	High	-24.1883, 31.0605	Apr 15, 2026, 10:28 AM	ranger12	Synced
Incident	An unidentified person was seen acting suspici...	Medium	-24.3400, 31.0682	Sep 23, 2026, 4:28 PM	ranger3	Synced
Incident	Unusual human tracks were found during a routi...	Low	-24.1789, 31.2381	Sep 19, 2026, 6:52 AM	ranger7	Synced
Incident	Unusual human tracks were found during a routi...	Low	-24.2229, 31.1397	Mar 16, 2026, 6:38 PM	ranger8	Synced
Incident	Unauthorized vehicle tracks were found inside t...	Medium	-24.0787, 31.1222	Apr 4, 2026, 1:21 AM	ranger6	Synced
Incident	Unusual human tracks were found during a routi...	Low	-24.0790, 31.1213	May 2, 2026, 7:36 AM	ranger2	Synced

< 1 ... 4 **5** 6 7 >

Submitting a new report

Tap "New Report", choose Sighting or Incident, then fill in the location, the date and time it happened, a description, and any photos.

The screenshot shows the 'Field Reports' interface. At the top, there's a 'NEW REPORT' button and an 'ALL REPORTS' button. Below that is a navigation bar with a back arrow, a '+' button, and a forward arrow. The 'Report Type' section has two radio buttons: 'Incident' (selected) and 'Sighting'. The 'Description' section has a text input field with a placeholder 'Describe what you observed...' and a character count '0 / 300'. The 'Incident Type' section has a dropdown menu labeled 'Select an incident type...'. The 'Severity' section has three radio buttons: 'Low', 'Medium', and 'High'. The 'When did this happen?' section has a date and time input field showing '28/09/2026 20:57'. The 'Location' section has a 'Use current location' button and a map showing 'Balule Nature Reserve', 'Klaserie Game Reserve', and 'Umbabat Nature Reserve'. Below the map are input fields for 'Latitude' and 'Longitude'. The 'Photos (optional)' section has a dashed border, a camera icon, and the text 'Add photos' and 'JPG or PNG, up to 5 photos, 10MB each'. At the bottom right is a 'Submit Report' button.

Once submitted, you can still edit or delete your own reports if they have not synced with the server, once they are synced you will no longer be able to edit or delete a field report.

Editing or deleting a report

Open a report from your list, and at the bottom press "Delete Draft" to delete the report, or change any info and press "Save Draft" to update your existing report.

The screenshot shows a row of buttons. On the left is a navigation bar with a back arrow, a '1' button, a '+' button, and a forward arrow. To the right of this are two buttons: a red 'Delete Draft' button and a blue 'Save Draft' button.

Data ingestion

Ingestion is how analysts bring historical data into the system from a CSV file

Preparing your file

The uploader expects a CSV with a fixed set of columns: `record_id`, `ingestion_timestamp`, `source_system`, `data_domain`, `event_type`, `payload_size_kb`, `priority_level`, `retry_count`, `is_encrypted`, `status`. The header row has to match this exactly. If a column is missing, renamed, or out of order, the file won't be accepted.

Uploading and previewing

Drag your CSV onto the upload area, or click to browse for it. Once it's accepted, you won't see the raw file, instead you get an editable table showing every row, so mistakes can be fixed on the spot before anything goes further.

Data Ingestion


Drop CSV here or click to browse
Accepts .csv only, header row must match the expected schema

No data loaded

Fixing problems before you submit

As you edit the table, any cell that doesn't match the expected type is flagged live with a red border and a short message explain what's wrong. Larger files are paged 50 rows at a time.

null Expected boolean, got "null"	"record_id" is missing	five Expected number, got "five"
---	---	--

Submitting

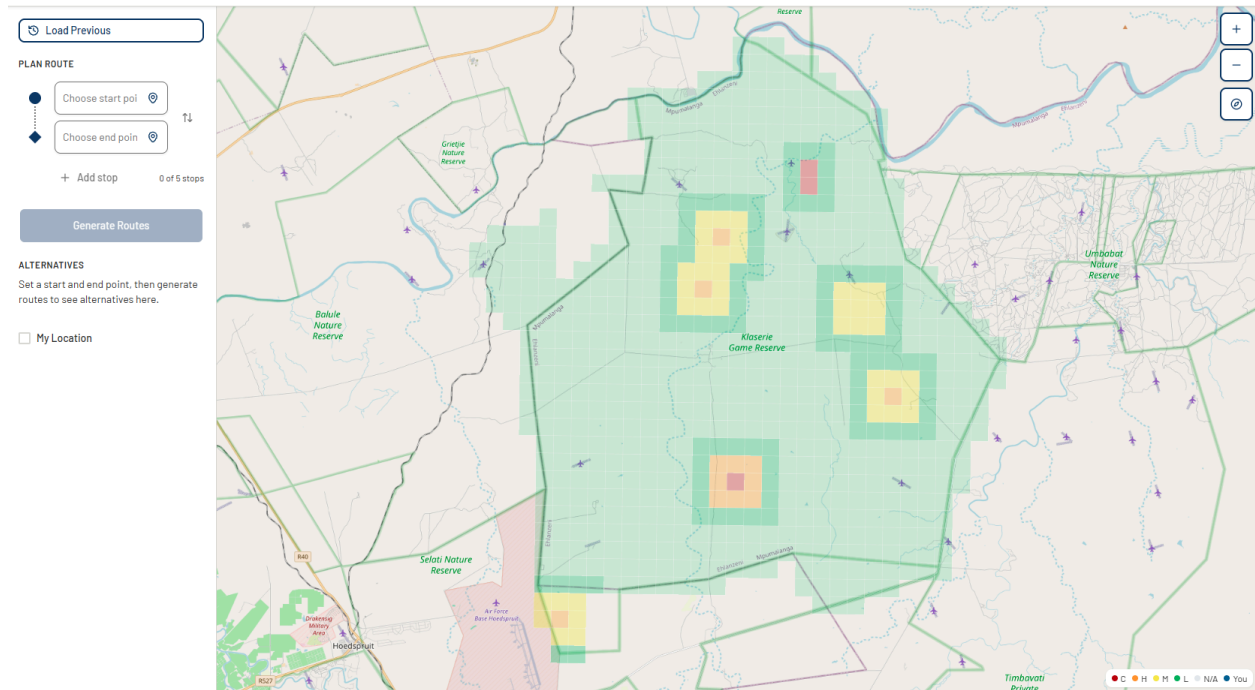
Once the table is clean, press "Submit".

Submit

If the batch goes through, you will see a completion and option to upload another file.

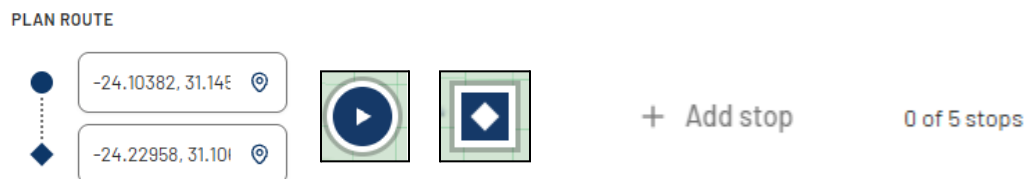
Patrol Planner

Open the Patrol Planner page to request suggested patrol routes for a park.



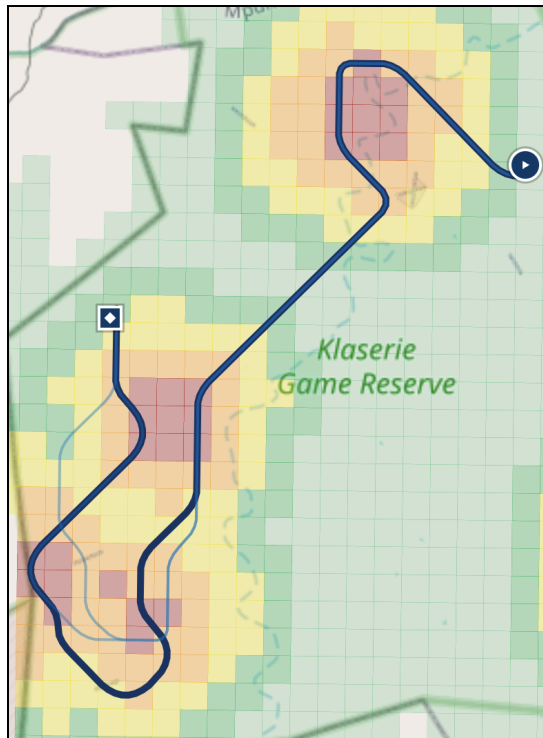
Requesting Routes

Set a start point and an end point for the patrol, either by typing coordinates or by clicking the pin button and tapping the two locations on the map. You may click the add stop button if needed, once done click the generate routes button.



Comparing routes

Up to three routes appear as lines on the map, along with cards showing each route's estimated distance, and risk coverage. Click a card to highlight that route on the map; the others dim. Press "Clear Routes" to discard the results and start over.



Clear Routes

ALTERNATIVES

Showing 2 of 3 alternatives: the remaining options retraced a route already shown

● Route A



Selected

Distance

67.4 km

Risk Coverage

100%

● Route B



Select

Distance

67.9 km

Risk Coverage

100%

If no route can be found, you'll see a message telling you to raise one of those limits and try again.

Viewing your location

Toggle the checkbox to see your location on the map, and a radius around it indicates the accuracy, as in the radius around the icon is indicative of your approximate location.

☐ My Location



Admin Tools

Admin accounts get an extra section covering user management and system oversight.

Account approvals

You can approve or reject a newly registered account

Pending Registrations				
Search pending registrations...				<button>Filters</button>
Username	Name	Role Claim	Created At	ACTIONS
ranger2	Ranger Pending	Ranger	25/07/2026	<button>Accept</button> <button>Reject</button>

Role Management

You can change a user's role

Role Management				
Search active users...				<button>Filters</button>
Username	Name	Email	Current Role	ASSIGN ROLE
ranger1	Ranger One	ranger1@sentinel.dev	Ranger	Ranger <button>Apply</button>
analyst1	Analyst One	analyst1@sentinel.dev	Analyst	Analyst <button>Apply</button>
liaison1	Liaison One	liaison1@sentinel.dev	Community Liaison	Community Liaison <button>Apply</button>

Delete Accounts

You can delete a user.

Delete Accounts				
Search active users...				<button>Filters</button>
Username	Name	Email	Current Role	ACTIONS
ranger1	Ranger One	ranger1@sentinel.dev	Ranger	<button>Delete</button>
analyst1	Analyst One	analyst1@sentinel.dev	Analyst	<button>Delete</button>
liaison1	Liaison One	liaison1@sentinel.dev	Community Liaison	<button>Delete</button>

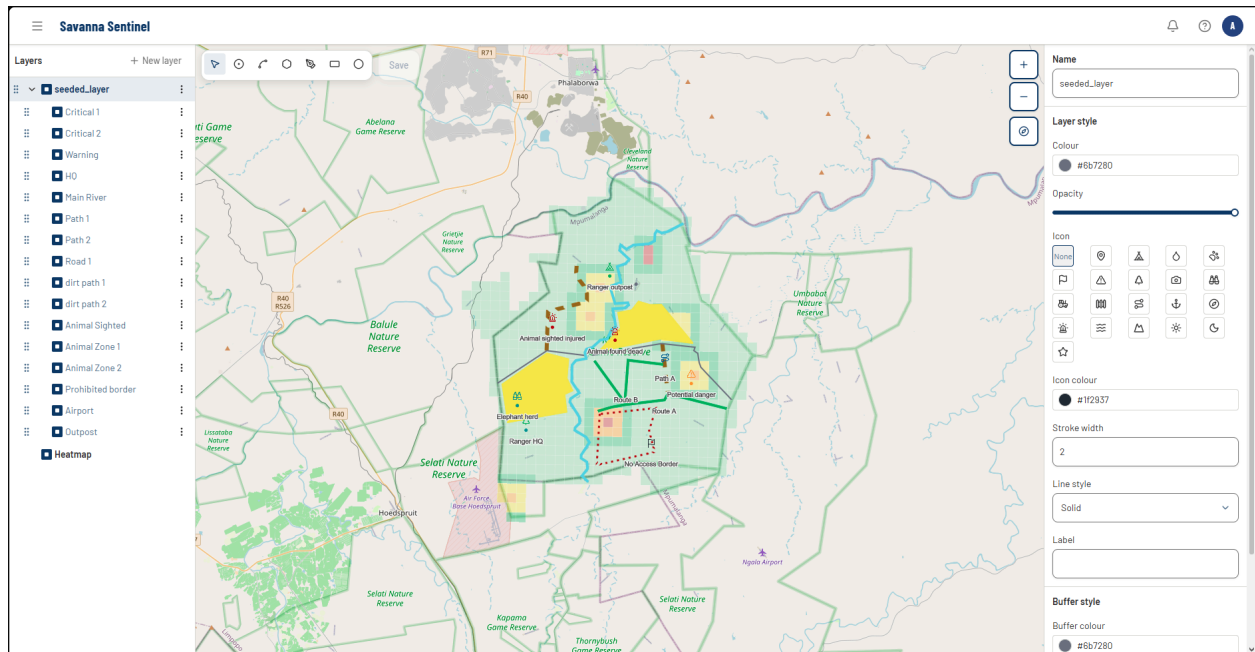
Audit log

Every admin action - activating a user, changing a role, deleting a user.

View Audit Logs					
ACTOR ID	ACTION	TARGET	TARGET ID	DETAILS	CREATED AT
bf5d34b6-5240-4857-94e4-129f575362cb	user.soft_deleted	user	bbc248b1-e75c-4274-b656-4bd59719093f	No details	2026-07-28T14:26:33.186899Z
bf5d34b6-5240-4857-94e4-129f575362cb	user.account_accepted	user	bbc248b1-e75c-4274-b656-4bd59719093f	No details	2026-07-28T14:26:25.272771Z
bf5d34b6-5240-4857-94e4-129f575362cb	user.role_changed	user	abc4910c-b7cb-4e69-8838-46393f753a8e	Role changed to ranger	2026-07-28T14:26:21.243115Z
bf5d34b6-5240-4857-94e4-129f575362cb	user.role_changed	user	abc4910c-b7cb-4e69-8838-46393f753a8e	Role changed to analyst	2026-07-28T14:26:17.653845Z
bf5d34b6-5240-4857-94e4-129f575362cb	user.role_changed	user	27c7c878-3dc9-4c5e-a143-193cfe833778	Role changed to analyst	2026-07-28T14:26:12.769763Z
< 1 >					

This log can't be edited or removed by anyone, including admins.

Workspace



In the workspace you can create, view, and toggle map layers.

+ New layer

Click this button in the left pane at the top left to create a new layer.



These are your drawing tools,

- Select - To drag around the map and select layers and points.
- Point - To create a point, click anywhere on the map.
- Line tool - To create a line.
- Polygon tool - To create a filled in polygon on the map.
- Free draw tool - To create either a free draw line or polygon.
- Rectangle/Circle tool - To create a respective shape.

The right pane

The right pane contains settings for a layer named "seeded_layer". It is divided into two main sections: "Layer style" and "Buffer".

Layer style

- Name:** seeded_layer
- Layer style:**
 - Colour:** #6b7280
 - Opacity:** A slider bar.
 - Icon:** A grid of 20 icons. The first icon, "None", is selected.
 - Icon colour:** #1f2937
 - Stroke width:** 2
 - Line style:** Solid (with a dropdown arrow)
 - Label:** An empty text input field.

Buffer

- Enable buffer:** ☒
- Distance (metres):** 750 (with a slider bar)
- Buffer colour:** #009193
- Buffer opacity:** A slider bar, with a [Reset](#) link.

Terrain Aware Intelligence

- Behaviour:** A button.

Modifications

- In effect:** ☒
- Edit geometry:** A button.

There are various inputs and fields you can access here to modify the points you've created on the map.

Rules

Rules shape the heatmap and patrol routes. Where rules overlap, even within one feature, the highest priority number wins. Risk and route rules are compared separately. Equal priorities all apply.

Compare priorities across features

☐ **Increase risk**

Cells here score higher on the heatmap.

☐ **Decrease risk**

Cells here score lower on the heatmap.

☐ **Prefer following**

Patrol routes are steered towards this.

☐ **Avoid following**

Patrol routes are steered away from this.



Increase risk

Cells here score higher on the heatmap.

Strength 50%



Buffer decay 0%



Turn on the buffer to use decay.

Priority

1

Each polygon can be assigned certain rules. Each rule has a strength associated with it, which determines how much of an effect it has on the Patrol Planner/Heatmap. The buffer decay is how much this strength will decrease as the distance from the feature in the buffer increases. Below follows a short explanation of each rule:

- Increase Risk - If a sighting or incident occurs in this area, the cells will gain an increased risk score.
- Decrease Risk - If a sighting or incident occurs in this area, the effect of the report will be decreased.
- Prefer following - The patrol planner will prefer following this feature when planning a route.
- Avoid Following - The patrol planner will avoid crossing this feature when planning a route.

Each rule has an associated priority. The priority of the feature becomes relevant when 2 features intersect. In this case the feature with higher priority will take effect and cancel out the lower priority. This can be used to make a river that cannot be crossed except on a higher priority line which acts as a bridge.